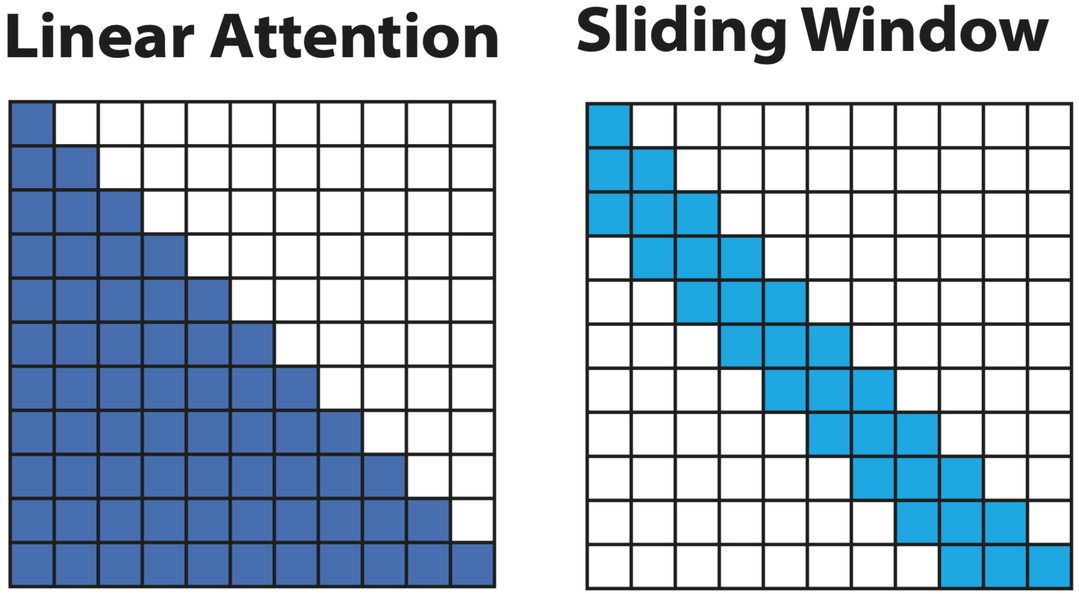
State-Space Models

Quadratic Attention

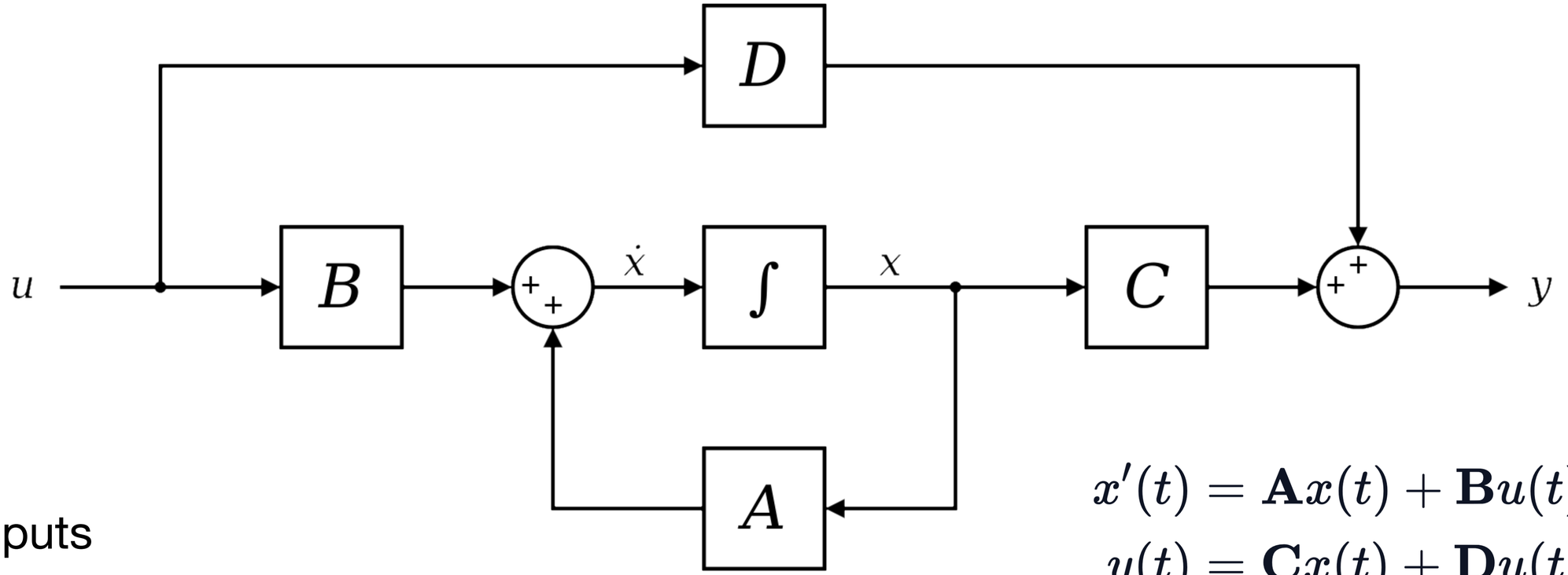
- Vanila attention has quadratic complexity

- It makes Transformers hard to apply to long contexts
- Especially in the era on reasoning

- Sliding window attention (and others) is a solution, but we can do better



State Space Models



$$x'(t) = \mathbf{A}x(t) + \mathbf{B}u(t)$$
$$y(t) = \mathbf{C}x(t) + \mathbf{D}u(t)$$

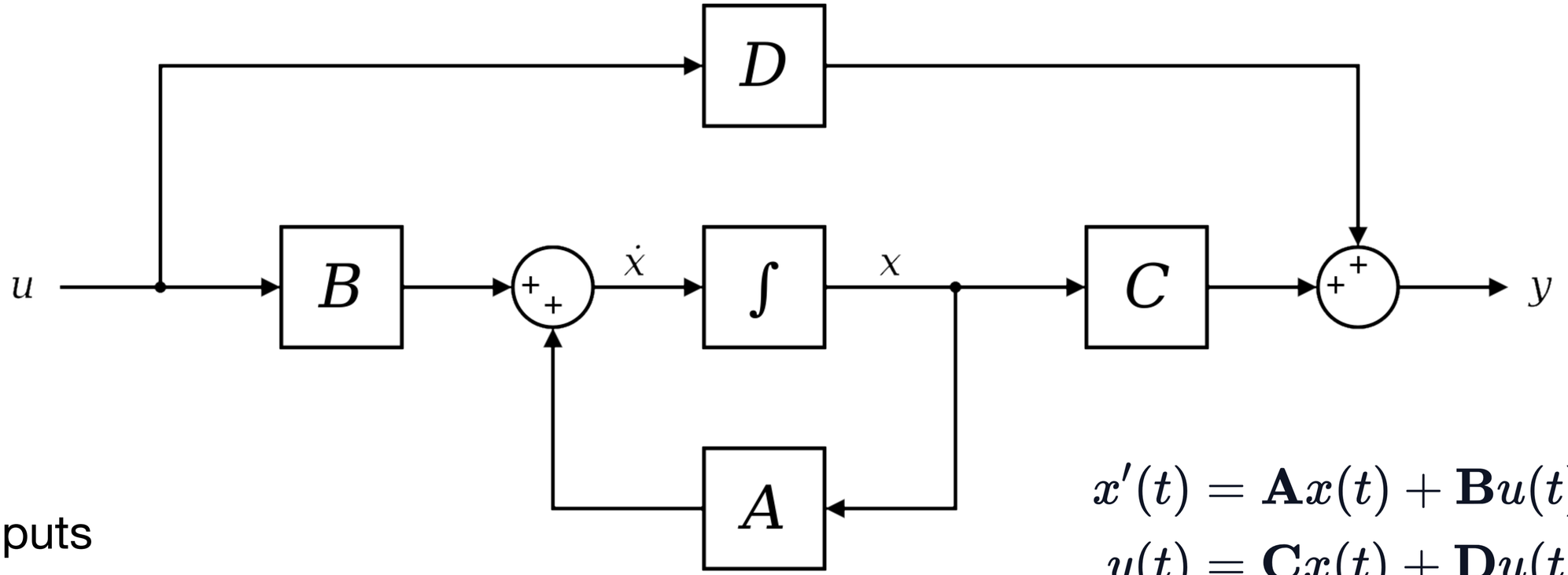
u is inputs

y is outputs

x is a hidden state (yes, like in RNNs)

$\dot{x}$ is the derivative

State Space Models



u is inputs

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$$x'(t) = \mathbf{A}x(t) + \mathbf{B}u(t)$$
$$y(t) = \mathbf{C}x(t) + \mathbf{D}u(t)$$

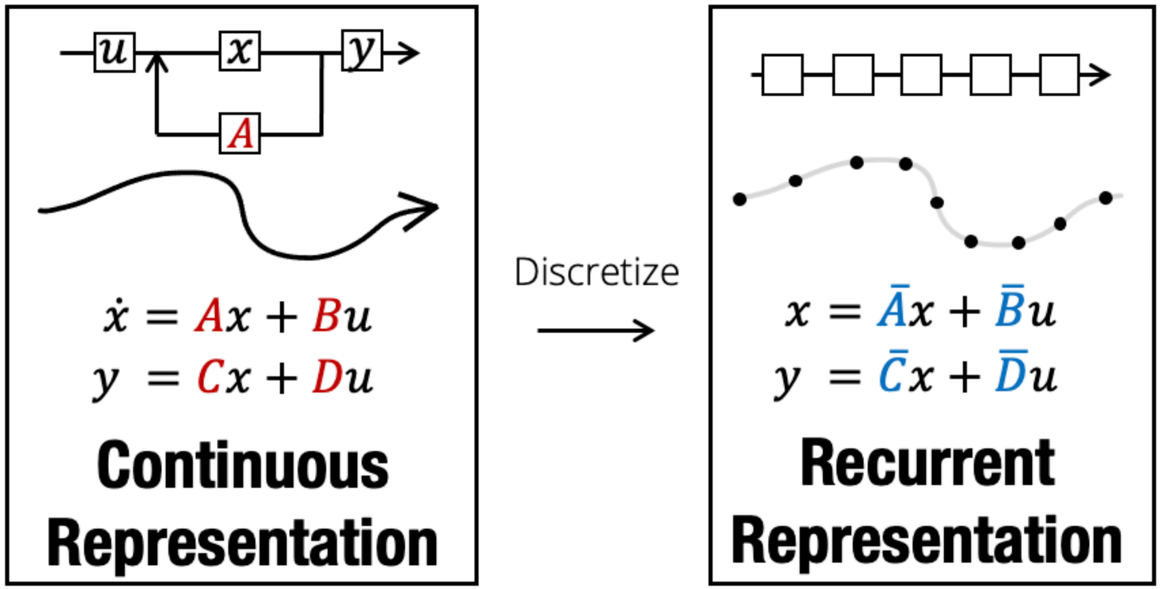
or

$$x' = \mathbf{A}x + \mathbf{B}u$$
$$y = \mathbf{C}x$$

State Space Models

This system is continuous and should be discretized

$$x' = \mathbf{A}x + \mathbf{B}u$$
$$y = \mathbf{C}x$$



Discretization

$$x' = \mathbf{A}x + \mathbf{B}u$$
$$y = \mathbf{C}x$$

Trapezoid method: $x_{n+1} - x_n = \frac{1}{2}\Delta(f(t_n) + f(t_{n+1}))$ $\Delta = t_{n+1} - t_n$

$$x_{n+1} = x_n + \frac{\Delta}{2}(\mathbf{A}x_n + \mathbf{B}u_n + \mathbf{A}x_{n+1} + \mathbf{B}u_{n+1})$$

Discretization

$$x' = \mathbf{A}x + \mathbf{B}u$$
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$$x_{n+1} = x_n + \frac{\Delta}{2}(\mathbf{A}x_n + \mathbf{B}u_n + \mathbf{A}x_{n+1} + \mathbf{B}u_{n+1})$$

$$\iff x_{n+1} - \frac{\Delta}{2}\mathbf{A}x_{n+1} = x_n + \frac{\Delta}{2}\mathbf{A}x_n + \frac{\Delta}{2}\mathbf{B}(u_{n+1} + u_n)$$

Discretization

$$x' = \mathbf{A}x + \mathbf{B}u$$
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Trapezoid method: $x_{n+1} - x_n = \frac{1}{2}\Delta(f(t_n) + f(t_{n+1}))$ $\Delta = t_{n+1} - t_n$

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$$(*) \iff (\mathbf{I} - \frac{\Delta}{2}\mathbf{A})x_{n+1} = (\mathbf{I} + \frac{\Delta}{2}\mathbf{A})x_n + \Delta\mathbf{B}u_{n+1}$$

$(*)\ u_{n+1} \overset{\Delta}{\simeq} u_n$

Discretization

$$x' = \mathbf{A}x + \mathbf{B}u$$
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Trapezoid method: $x_{n+1} - x_n = \frac{1}{2}\Delta(f(t_n) + f(t_{n+1}))$ $\Delta = t_{n+1} - t_n$

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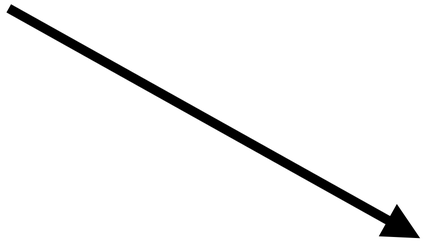
$$(*) \iff (\mathbf{I} - \frac{\Delta}{2}\mathbf{A})x_{n+1} = (\mathbf{I} + \frac{\Delta}{2}\mathbf{A})x_n + \Delta\mathbf{B}u_{n+1}$$
$$(*)\ u_{n+1} \overset{\Delta}{\simeq} u_n \iff x_{n+1} = (\mathbf{I} - \frac{\Delta}{2}\mathbf{A})^{-1}(\mathbf{I} + \frac{\Delta}{2}\mathbf{A})x_n + (\mathbf{I} - \frac{\Delta}{2}\mathbf{A})^{-1}\Delta\mathbf{B}u_{n+1}$$

Discretization

$$
\begin{aligned}
x' &= \mathbf{A}x + \mathbf{B}u \\
y &= \mathbf{C}x
\end{aligned}
$$



$$
\begin{aligned}
\bar{\mathbf{A}} &= (\mathbf{I} - \frac{\Delta}{2}\mathbf{A})^{-1}(\mathbf{I} + \frac{\Delta}{2}\mathbf{A}) \\
\bar{\mathbf{B}} &= (\mathbf{I} - \frac{\Delta}{2}\mathbf{A})^{-1}\Delta\mathbf{B} \\
\bar{\mathbf{C}} &= \mathbf{C}
\end{aligned}
$$

$$
\begin{aligned}
x_k &= \bar{\mathbf{A}}x_{k-1} + \bar{\mathbf{B}}u_k \\
y_k &= \bar{\mathbf{C}}x_k
\end{aligned}
$$

Discretization

$$x' = \mathbf{A}x + \mathbf{B}u$$
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$$\bar{\mathbf{B}} = (\mathbf{I} - \frac{\Delta}{2}\mathbf{A})^{-1}\Delta\mathbf{B}$$
$$\bar{\mathbf{C}} = \mathbf{C}$$

$$x_k = \bar{\mathbf{A}}x_{k-1} + \bar{\mathbf{B}}u_k$$
$$y_k = \bar{\mathbf{C}}x_k$$

This is called **S4** (Structured State Space Sequence model)

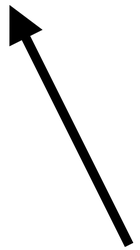
Recursive view

SSM

$$x_k = \bar{\mathbf{A}} x_{k-1} + \bar{\mathbf{B}} u_k$$
$$y_k = \bar{\mathbf{C}} x_k$$

RNN

$$h_t = \tanh(W_{hh} h_{t-1} + W_{xh} x_t)$$
$$y_t = W_{hy} h_t$$



Does not have non-linearity!

Convolutive view of an SSM

$$x_k = \bar{\mathbf{A}} x_{k-1} + \bar{\mathbf{B}} u_k$$
$$y_k = \bar{\mathbf{C}} x_k$$

Step 1: $x_0 = \bar{\mathbf{B}} u_0$

Step 2: $x_1 = \bar{\mathbf{A}} x_0 + \bar{\mathbf{B}} u_1 = \bar{\mathbf{A}}\bar{\mathbf{B}} u_0 + \bar{\mathbf{B}} u_1$

Step 3: $x_2 = \bar{\mathbf{A}} x_1 + \bar{\mathbf{B}} u_2 = \bar{\mathbf{A}}(\bar{\mathbf{A}}\bar{\mathbf{B}} u_0 + \bar{\mathbf{B}} u_1) + \bar{\mathbf{B}} u_2 = \bar{\mathbf{A}}^2\bar{\mathbf{B}} u_0 + \bar{\mathbf{A}}\bar{\mathbf{B}} u_1 + \bar{\mathbf{B}} u_2$

x_k is a linear function of $(u_0, u_1, ...u_k)$

Convolutive view of an SSM

$$x_k = \bar{\mathbf{A}} x_{k-1} + \bar{\mathbf{B}} u_k$$
$$y_k = \bar{\mathbf{C}} x_k$$

Step 1: $x_0 = \bar{\mathbf{B}} u_0$

Step 2: $x_1 = \bar{\mathbf{A}} x_0 + \bar{\mathbf{B}} u_1 = \bar{\mathbf{A}}\bar{\mathbf{B}} u_0 + \bar{\mathbf{B}} u_1$

Step 3: $x_2 = \bar{\mathbf{A}} x_1 + \bar{\mathbf{B}} u_2 = \bar{\mathbf{A}}(\bar{\mathbf{A}}\bar{\mathbf{B}} u_0 + \bar{\mathbf{B}} u_1) + \bar{\mathbf{B}} u_2 = \bar{\mathbf{A}}^2\bar{\mathbf{B}} u_0 + \bar{\mathbf{A}}\bar{\mathbf{B}} u_1 + \bar{\mathbf{B}} u_2$

Step 1: $y_0 = \bar{\mathbf{C}} x_0 = \bar{\mathbf{C}}\bar{\mathbf{B}} u_0$

Step 2: $y_1 = \bar{\mathbf{C}} x_1 = \bar{\mathbf{C}}(\bar{\mathbf{A}}\bar{\mathbf{B}} u_0 + \bar{\mathbf{B}} u_1) = \bar{\mathbf{C}}\bar{\mathbf{A}}\bar{\mathbf{B}} u_0 + \bar{\mathbf{C}}\bar{\mathbf{B}} u_1$

Step 3: $y_2 = \bar{\mathbf{C}} x_2 = \bar{\mathbf{C}}(\bar{\mathbf{A}}^2\bar{\mathbf{B}} u_0 + \bar{\mathbf{A}}\bar{\mathbf{B}} u_1 + \bar{\mathbf{B}} u_2) = \bar{\mathbf{C}}\bar{\mathbf{A}}^2\bar{\mathbf{B}} u_0 + \bar{\mathbf{C}}\bar{\mathbf{A}}\bar{\mathbf{B}} u_1 + \bar{\mathbf{C}}\bar{\mathbf{B}} u_2$

$$\bar{\mathbf{K}}_k = (\bar{\mathbf{C}}\bar{\mathbf{B}}, \bar{\mathbf{C}}\bar{\mathbf{A}}\bar{\mathbf{B}}, ..., \bar{\mathbf{C}}\bar{\mathbf{A}}^k\bar{\mathbf{B}})$$

When to use each view

Continuous view:

✓ Handles continuous data (audio signals, time series)

✗ Extremely slow for both training and inference

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✓ Efficient inference
✗ Slow learning (lack of parallelism)
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✗ Slow learning (lack of parallelism)
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Convolutional view:

✓ Efficient (parallelizable) training
✗ Slow in autoregressive contexts (must recalculate entire input for each new data point)
✗ Fixed context size

Slow powering on A

$$\bar{\mathbf{K}}_k = (\bar{\mathbf{C}}\bar{\mathbf{B}}, \bar{\mathbf{C}}\bar{\mathbf{A}}\bar{\mathbf{B}}, ..., \bar{\mathbf{C}}\bar{\mathbf{A}}^k\bar{\mathbf{B}})$$

- We need to compute $\bar{\mathbf{A}}^k$ fast
- Easiest way is to make $\bar{\mathbf{A}}^k$ diagonal (class of Normal matrices)

$$\bar{\mathbf{A}}^{\mathbf{k}} = \begin{bmatrix} \lambda_1^k & 0 & \cdots & 0 \\ 0 & \lambda_2^k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n^k \end{bmatrix}$$

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- However, initialization matters a lot here.

HiPPO

High-Order Polynomial Projection Operator

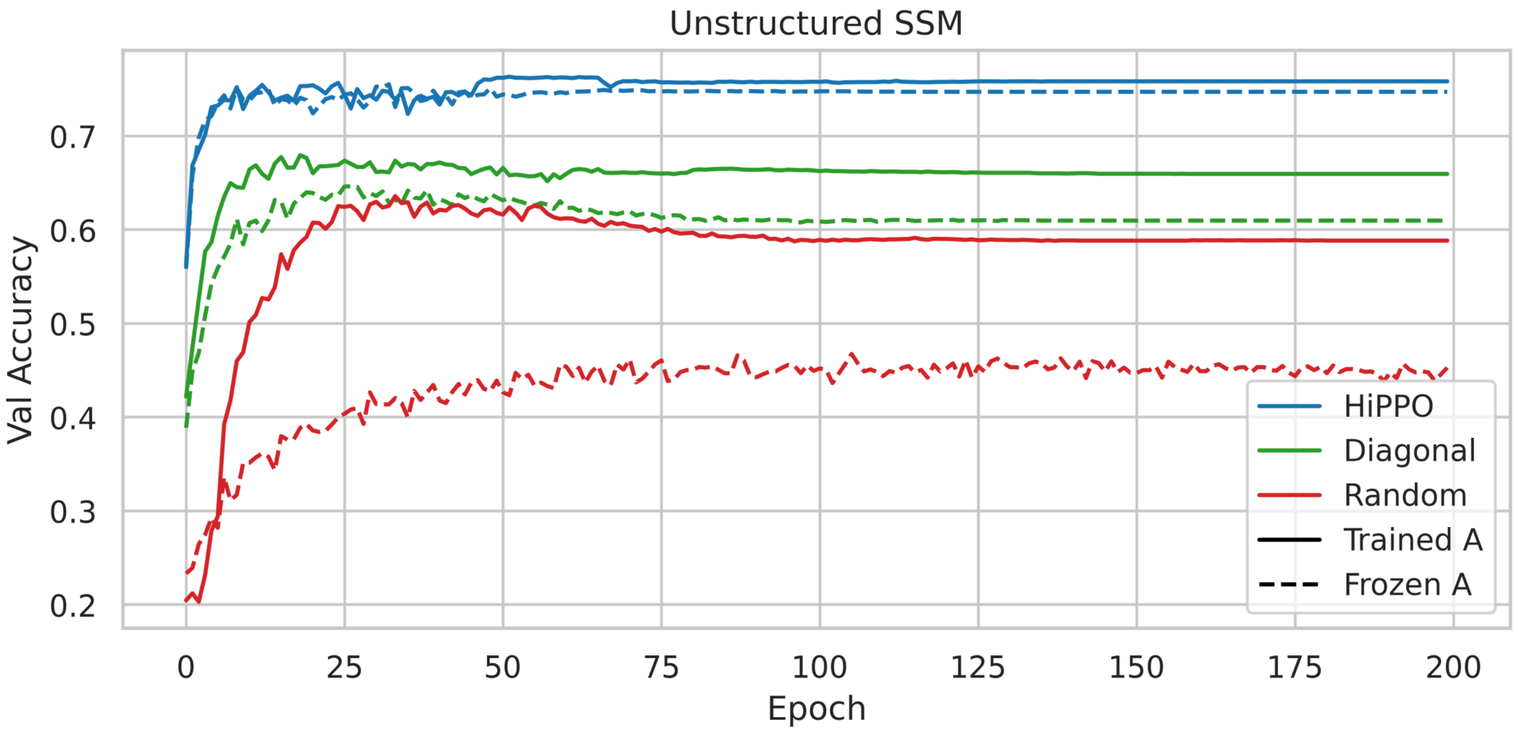
- Inits $\bar{\mathbf{A}}^k$ with NPLR (Normal Plus Low Rank) matrix
- Boosts quality from 60% to 98% on the MNIST sequential benchmark

$$\mathbf{A} = \begin{bmatrix} 1 & & & & & & & & \\ -1 & 2 & & & & & & & \\ 1 & -3 & 3 & & & & & & \\ -1 & 3 & -5 & 4 & & & & & \\ 1 & -3 & 5 & -7 & 5 & & & & \\ -1 & 3 & -5 & 7 & -9 & 6 & & & \\ 1 & -3 & 5 & -7 & 9 & -11 & 7 & & \\ -1 & 3 & -5 & 7 & -9 & 11 & -13 & 8 & \\ \vdots & & & & & & & & \ddots \end{bmatrix}$$

$$\Rightarrow \mathbf{A}_{nk} = \begin{cases} (-1)^{n-k}(2k+1) & n > k \\ k+1 & n = k \\ 0 & n < k \end{cases}$$

HiPPO

High-Order Polynomial Projection Operator



CIFAR-10 classification accuracy

HiPPO

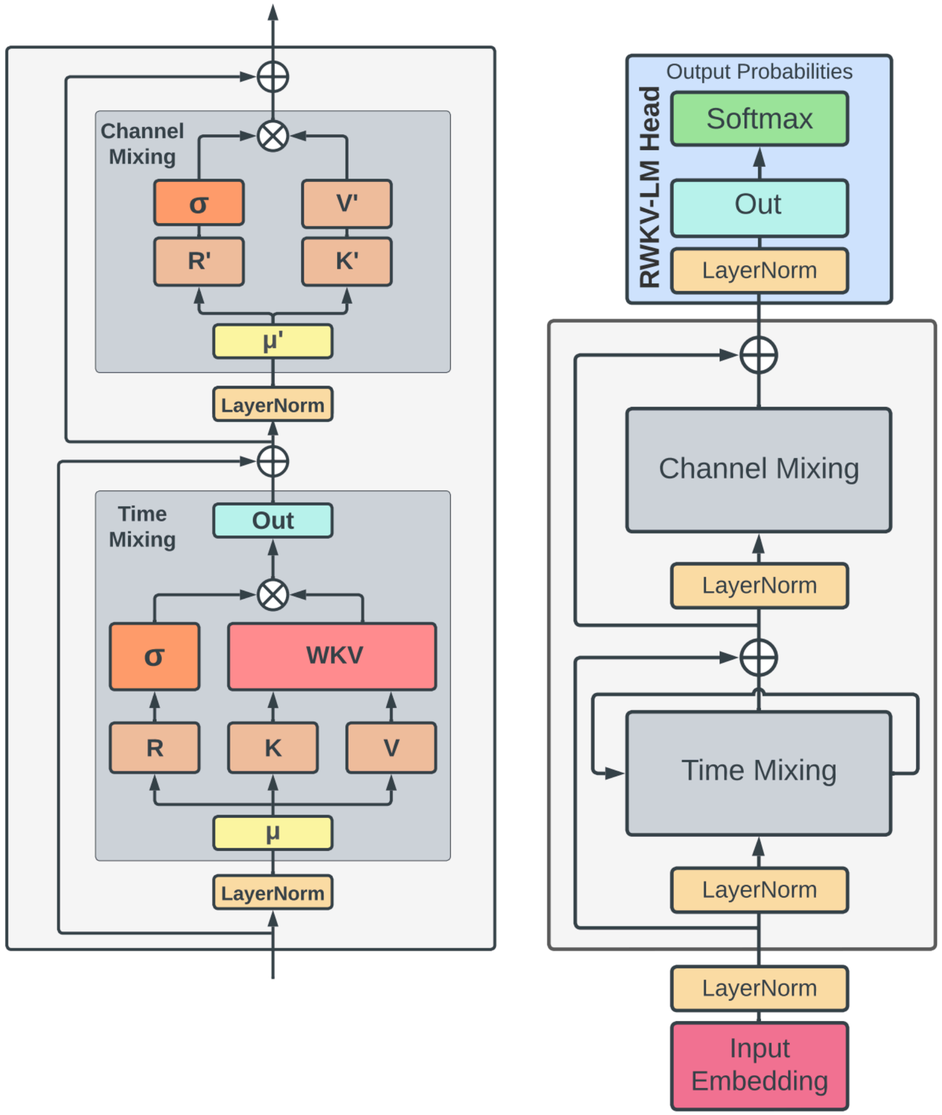
High-Order Polynomial Projection Operator

Model	ListOps	Text	Retrieval	Image	Pathfinder	Path-X	Avg
Transformer	36.37	64.27	57.46	42.44	71.40	✗	53.66
Reformer	<u>37.27</u>	56.10	53.40	38.07	68.50	✗	50.56
BigBird	36.05	64.02	59.29	40.83	74.87	✗	54.17
Linear Trans.	16.13	<u>65.90</u>	53.09	42.34	75.30	✗	50.46
Performer	18.01	65.40	53.82	42.77	77.05	✗	51.18
FNet	35.33	65.11	59.61	38.67	<u>77.80</u>	✗	54.42
Nyströmformer	37.15	65.52	<u>79.56</u>	41.58	70.94	✗	57.46
Luna-256	37.25	64.57	79.29	<u>47.38</u>	77.72	✗	<u>59.37</u>
S4	**59.60**	**86.82**	**90.90**	**88.65**	**94.20**	**96.35**	**86.09**

Long Range Arena benchmark

RWKV

Transformer layer is splitted in
Time-mixing and **Channel-mixing**



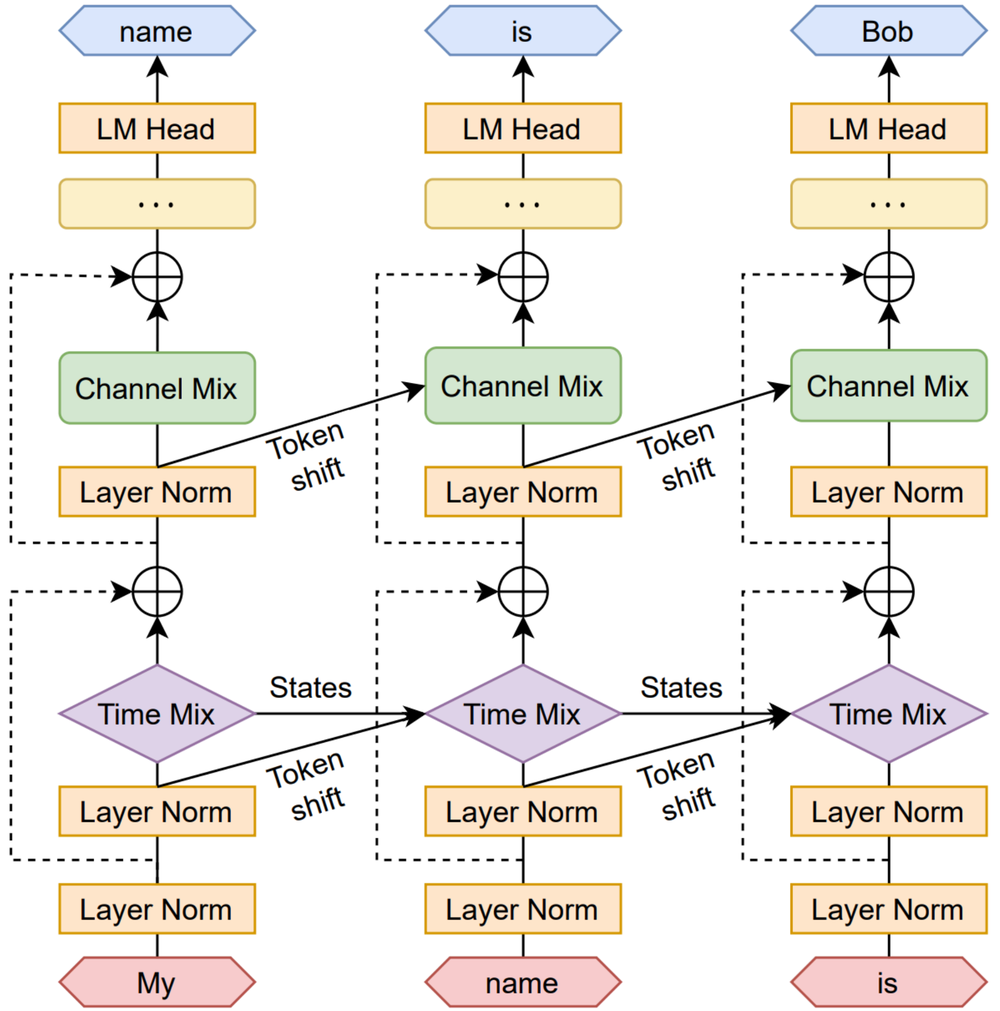
RWKV

Time-mixing

$$r_t = W_r \cdot (\mu_r \odot x_t + (1 - \mu_r) \odot x_{t-1})$$
$$k_t = W_k \cdot (\mu_k \odot x_t + (1 - \mu_k) \odot x_{t-1})$$
$$v_t = W_v \cdot (\mu_v \odot x_t + (1 - \mu_v) \odot x_{t-1})$$

$$wkv_t = \frac{\sum_{i=1}^{t-1} e^{-(t-1-i)w+k_i} \odot v_i + e^{u+k_t} \odot v_t}{\sum_{i=1}^{t-1} e^{-(t-1-i)w+k_i} + e^{u+k_t}}$$

$$o_t = W_o \cdot (\sigma(r_t) \odot wkv_t)$$

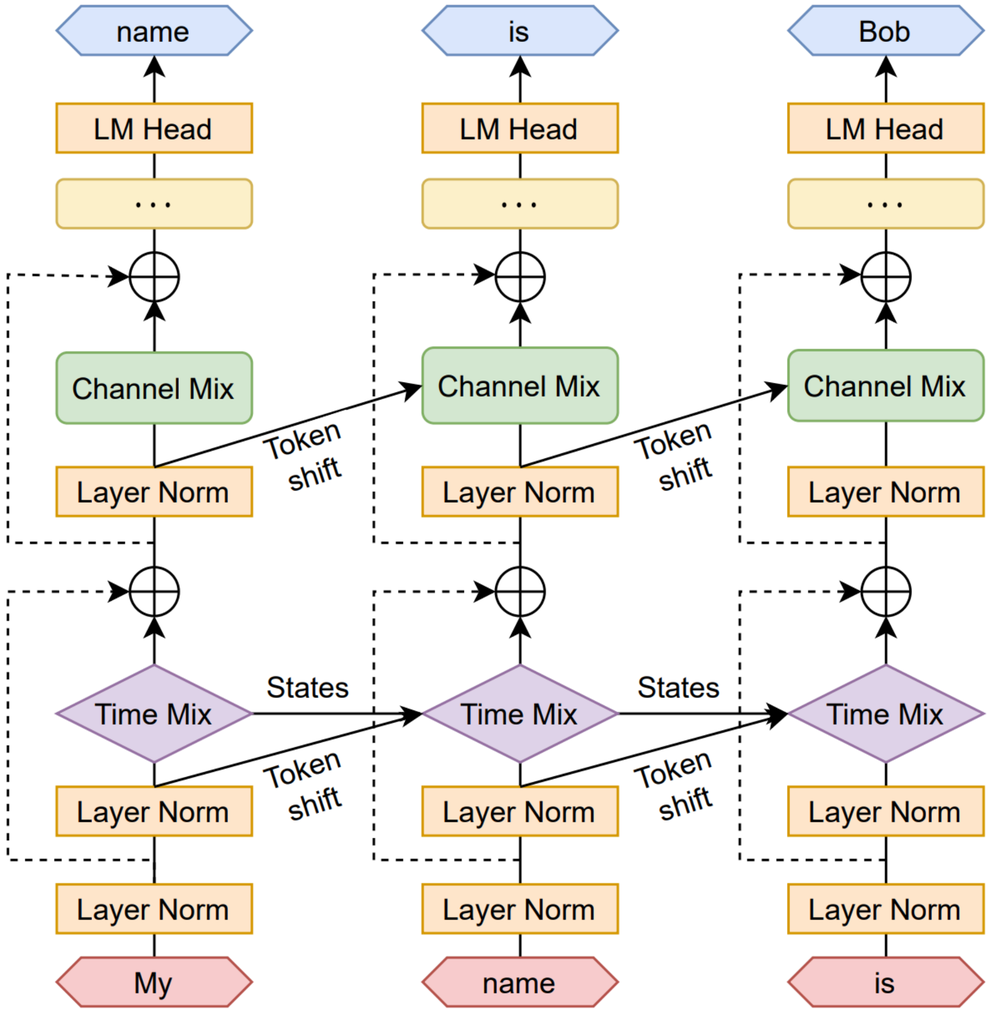


RWKV

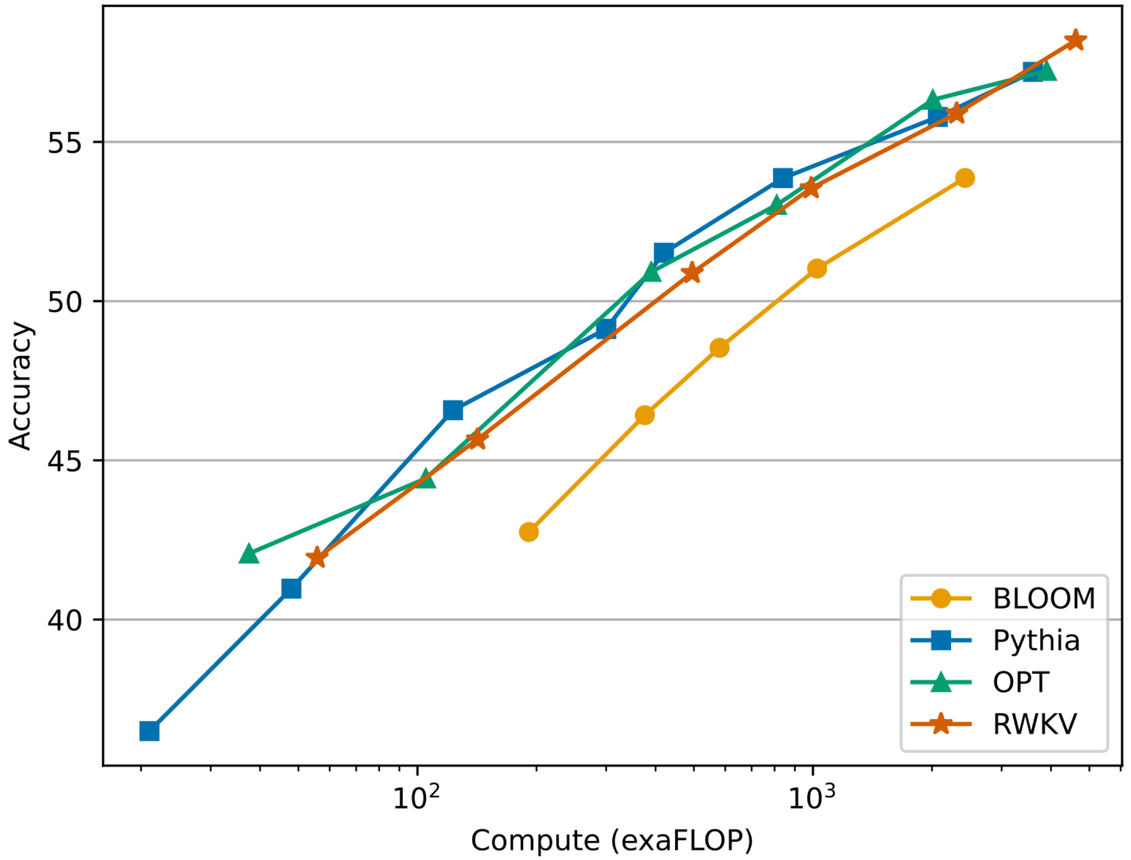
Channel-mixing

$$r'_t = W'_r \cdot (\mu'_r \odot x_t + (1 - \mu'_r) \odot x_{t-1})$$
$$k'_t = W'_k \cdot (\mu'_k \odot x_t + (1 - \mu'_k) \odot x_{t-1})$$

$$o'_t = \sigma(r'_t) \odot (W'_v \cdot \max(k'_t, 0)^2)$$



RWKV: results



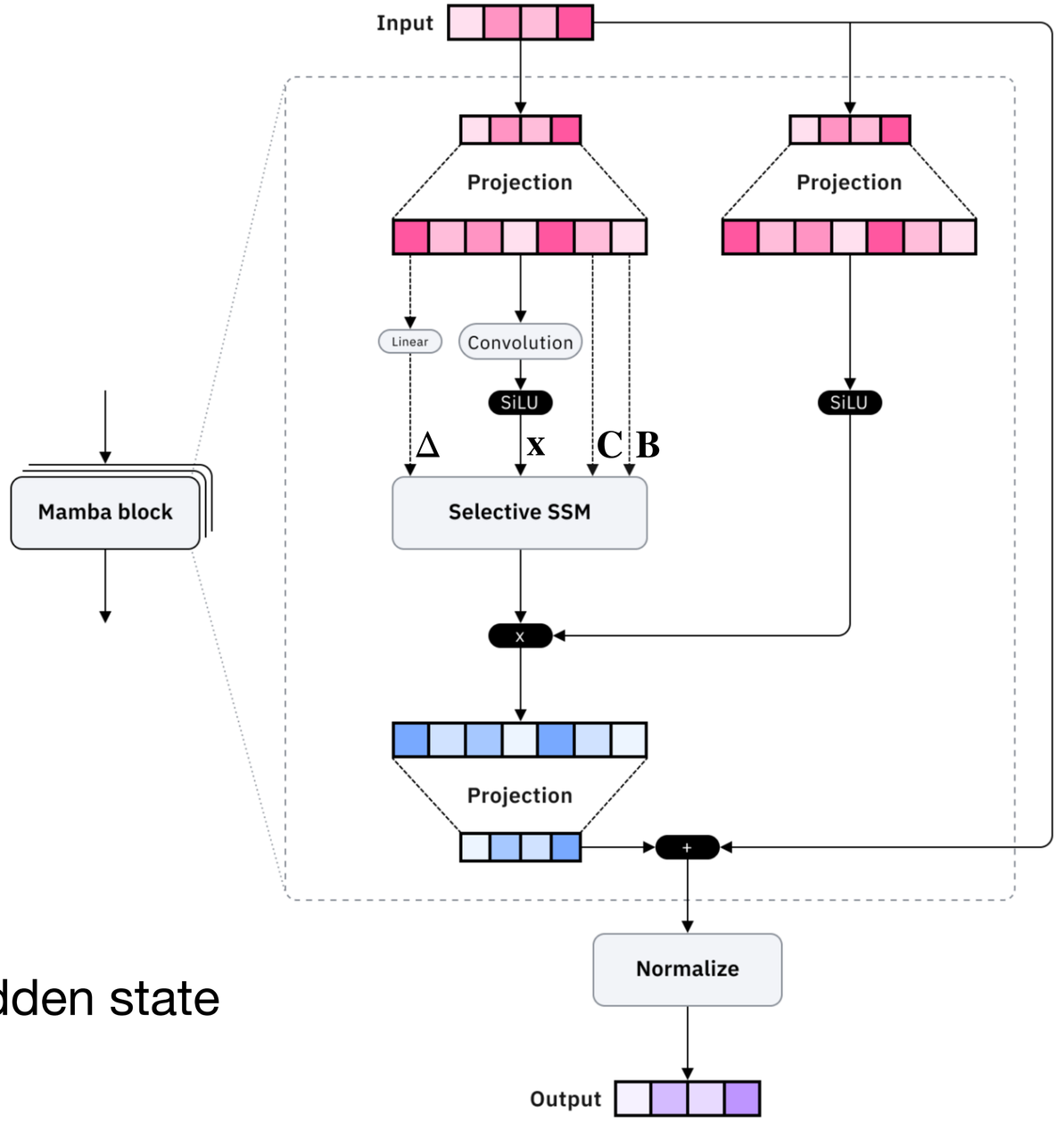
Mamba

Selective State Space model

$$\mathbf{A} \leftarrow \text{Parameter}$$
$$\mathbf{B} \leftarrow \text{Linear}_{\text{B}}(\mathbf{x})$$
$$\mathbf{C} \leftarrow \text{Linear}_{\text{C}}(\mathbf{x})$$
$$\Delta \leftarrow \text{Linear}_{1}(\mathbf{x})$$

$$\mathbf{x}_t = \bar{\mathbf{A}}\mathbf{x}_{t-1} + \bar{\mathbf{B}}\mathbf{u}$$
$$\mathbf{y} = \bar{\mathbf{C}}\mathbf{x}_t$$

Δ – step size (how much to forget)
$\mathbf{B}$ – how the current token updates the hidden state
$\mathbf{C}$ – how the context to affects the output



Mamba problem

Selective State Space model

$$\mathbf{A} \leftarrow \text{Parameter}$$
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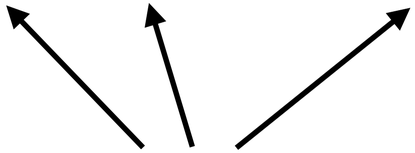


Depend on current hidden state!
Convolutional view can't applied.

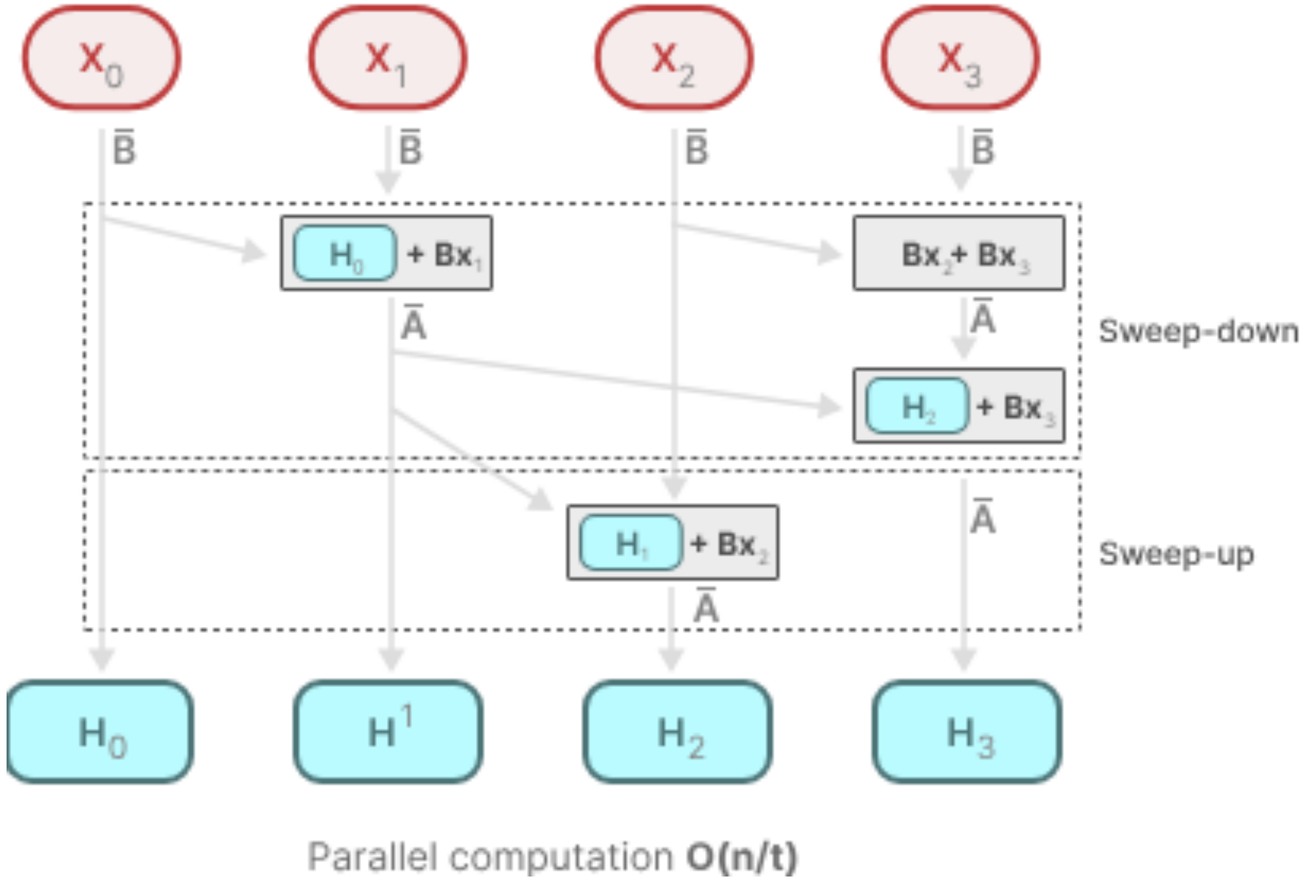
Parallel prefix sum

Computes prefix sums in parallel

$$\mathbf{A} \leftarrow \text{Parameter}$$
$$\mathbf{B} \leftarrow \text{Linear}_{\text{B}}(\mathbf{x})$$
$$\mathbf{C} \leftarrow \text{Linear}_{\text{C}}(\mathbf{x})$$
$$\Delta \leftarrow \text{Linear}_{1}(\mathbf{x})$$

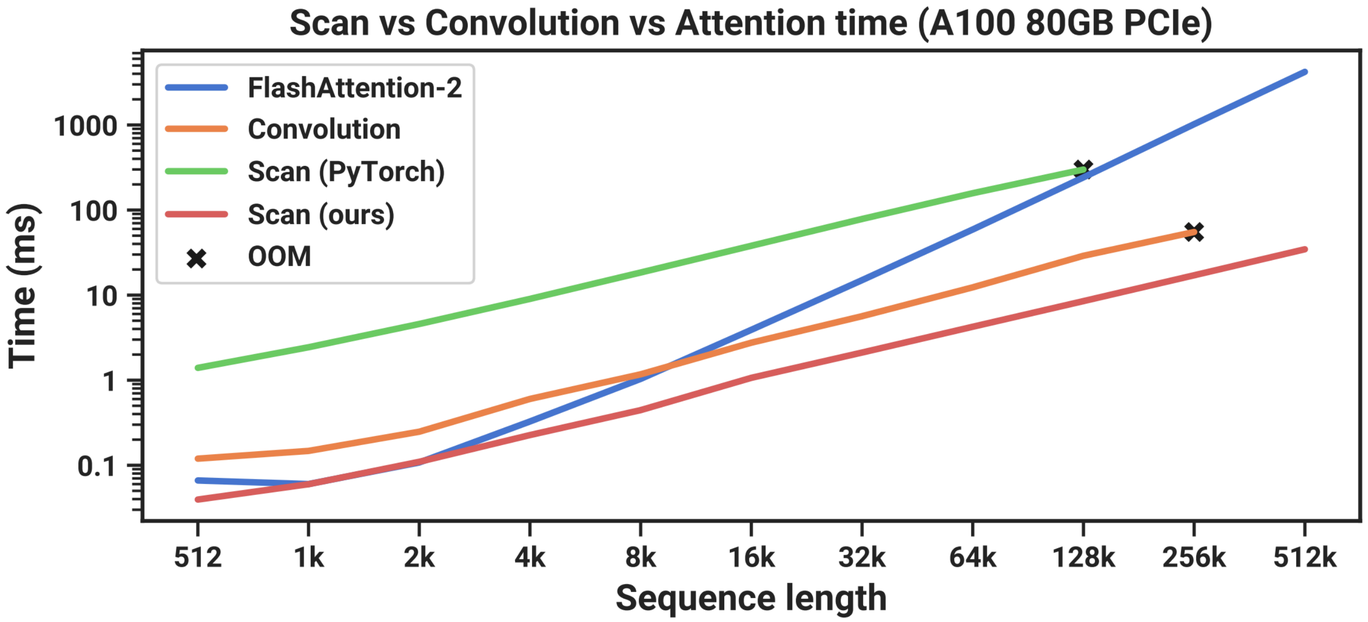
$$\mathbf{x}_t = \bar{\mathbf{A}}\mathbf{x}_{t-1} + \bar{\mathbf{B}}\mathbf{u}$$
$$\mathbf{y} = \bar{\mathbf{C}}\mathbf{x}_t$$

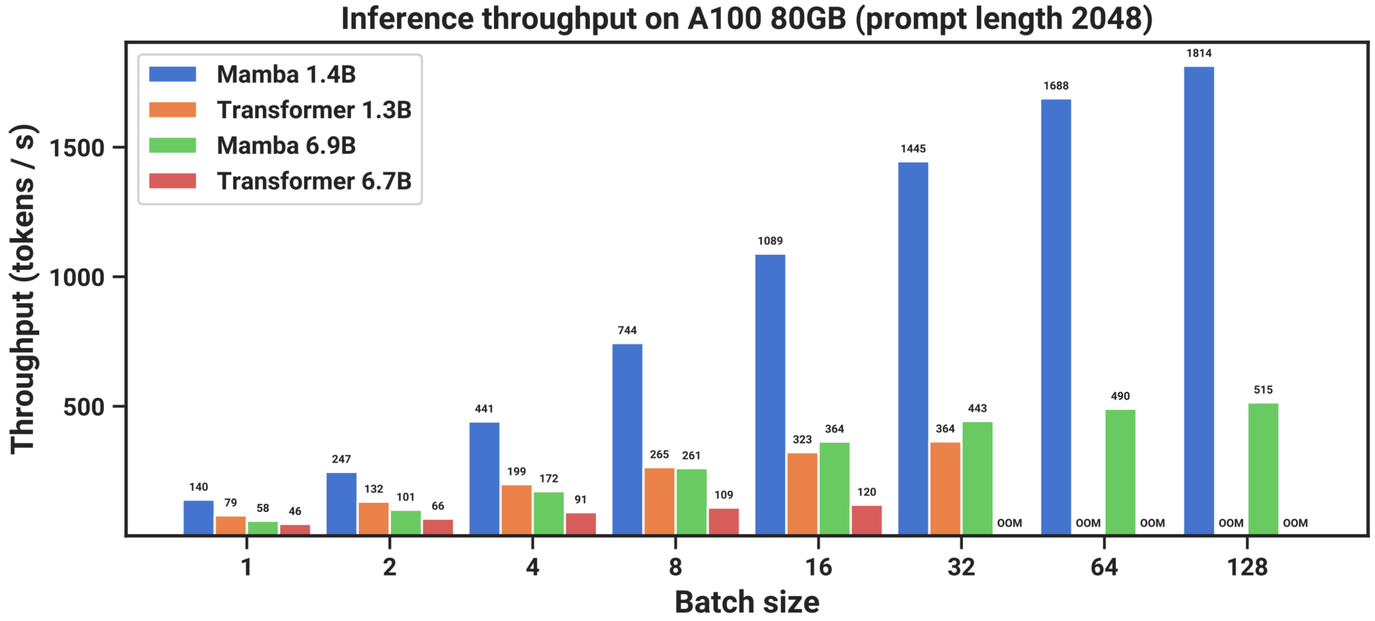
$$\mathbf{y} = \bar{\mathbf{C}}\bar{\mathbf{A}}^2\bar{\mathbf{B}}u_0 + \bar{\mathbf{C}}\bar{\mathbf{A}}\bar{\mathbf{B}}u_1 + \bar{\mathbf{C}}\bar{\mathbf{B}}u_2$$



Parallel computation **O(n/t)**

Mamba results





Mamba results